



Theming

After your development environment is fully set up, you can start building the skeleton of your theme module. In this chapter, you will discover how to:

- Enable/disable the Website Builder's standard options and templates.
- Define the colors and fonts to use for your design.
- Get the most out of Bootstrap variables.
- Add custom styles and JavaScript.

Theme module

Odoo comes with a default theme that provides minimal structure and layout. When you create a new theme, you are extending the default theme.

Remember to add the directory containing your module to the `addons-path` command-line argument when running Odoo in your development environment.

Technical naming

The first step is to create a new directory.

```
website_airproof
```

Note

Prefix it with `website_` and use only lowercase ASCII alphanumeric characters and underscores.

In this documentation, we will use **Airproof** (a fictional project) as an example.

File structure

Themes are packaged like any Odoo module. Even if you are designing a basic website, you need to package its theme like a module.

```

├─ static
│   ├── description
│   ├── fonts
│   ├── lib
│   ├── image_shapes // Shapes for images
│   ├── shapes // Shapes for background
│   └── src
│       ├── img
│       │   ├── content // For those used in the pages of your website
│       │   └── wbuilder // For those used in the builder
│       ├── js
│       ├── scss // Theme specific styles
│       ├── snippets // Custom snippets
│       └── website_builder // Options for the Website Builder
├─ views
├─ __init__.py
└─ __manifest__.py

```

Folder	Description
data	Presets, menus, pages, images, shapes, ... (*.xml)
i18n	Translations (*.po , *.pot)
lib	External libraries (*.js)
static	Custom assets (*.jpg , *.gif , *.png , *.svg , *.pdf , *.scss , *.js)
views	Custom views and templates (*.xml)

Initialization

An Odoo module is also a Python package with a `__init__.py` file containing import instructions for various Python files in the module. This file can remain empty for now.

Declaration

An Odoo module is declared by its manifest file. This file declares a Python package as an Odoo module and specifies the module's metadata. It must at least contain the `name` field, which is always required. It usually contains much more information.

```

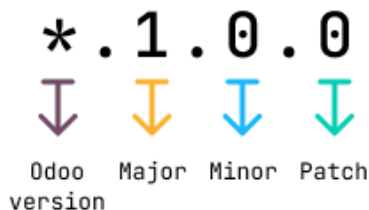
'description': '...',
'category': 'Website/Theme',
'version': '19.0.0',
'author': '...',
'license': '...',
'depends': ['website'],
'data': [
    # ...
],
'assets': {
    # ...
},
}

```

Field	Description
name	Human-readable name of the module (required)
description	Extended description of the module, in reStructuredText
category	Classification category within Odoo
version	Odoo version this module is addressing
author	Name of the module author
license	By default, we use the LGPL-3 license. More information in the module manifest page.
depends	Odoo modules must be loaded before this one, either because this module uses features they create or because it alters resources they define
data	List of XML files
assets	List of SCSS and JS files

Note

- To create a website theme, you only need to install the Website app. If you need other apps (Blogs, Events, eCommerce, ...), you can also add them.
- Odoo version and major number are mandatory. However, patch number is optional. If you want to specify the required version of Odoo to run your module, you should use a five arguments structure, using the first two arguments to indicate your current Odoo version (`* = 19.0`).



Example: `19.0.1.0.0`

`odoo_major.odoo_minor.module_major.module_minor.module_patch`

Warning

Automated file inclusion using wildcard notations (ex.: `/myfolder/*.scss`) doesn't work in Odoo SaaS databases. In this case, include each file manually in the manifest.

Default options

First, try to construct your theme by using Odoo's default options. This ensures two things:

1. You do not re-invent something which already exists. For example, as Odoo provides an option to add a border on the footer, you shouldn't recode it yourself. Instead, enable the default option first, then extend it if needed.
2. The user can still use all of Odoo's features with your theme. For example, if you recode the border on the footer, you may break the default option or make it useless, giving the user a bad experience. Also, your recode might not work as well as the default option, as other Odoo features may rely on it.

- Do not use tabs.
- Never mix spaces and tabs.

See also

[Odoo coding guidelines](#)

Odoo variables

Odoo declares many CSS rules, most entirely customizable by overriding the related SCSS variables. To do so, create a *primary_variables.scss* file and add it to the `_assets_primary_variables` bundle.

Declaration

/website_airproof/__manifest__.py

```
'assets': {
    'web._assets_primary_variables': [
        'website_airproof/static/src/scss/primary_variables.scss'
    ],
},
```

By reading the source code, variables related to options are easily noticeable.

```
<BuilderSelectedItem
    id="'...'"
    title.translate="..."
    actionParam="[
        {
            action: 'websiteConfig',
            actionParam: {
                views: [...],
                vars: { 'header-template': 'sidebar' },
                checkVars: false,
            }
        }
    ]">
```

These variables can be overridden through the `$o-website-value-palettes` map, for example.

Global

Declaration

```
/website_airproof/static/src/scss/primary_variables.scss
```

```
$o-website-values-palettes: (  
  (  
    // Templates  
    // Colors  
    // Fonts  
    // Buttons  
    // ...  
  ),  
);
```

Tip

That file must only contain definitions and overrides of SCSS variables and mixins.

See also

[Primary variables SCSS](#) ↗

Fonts

You can embed any font on your website. The Website Builder automatically makes them available in the font selector.

Declaration

```
/website_airproof/static/src/scss/primary_variables.scss
```

```
$o-theme-font-configs: (  
  <font-name>: (  
    'family': <CSS font family list>,
```

```

        ...,
    ),
    ...,
)
)

```

Use

/website_airproof/static/src/scss/primary_variables.scss

```

$-website-values-palettes: (
  (
    'font':                '<font-name>',
    'headings-font':       '<font-name>',
    'navbar-font':         '<font-name>',
    'buttons-font':        '<font-name>',
  ),
);

```

Google fonts

/website_airproof/static/src/scss/primary_variables.scss

```

$-theme-font-configs: (
  'Poppins': (
    'family':                ('Poppins', sans-serif),
    'url':                   'Poppins:400,500',
    'properties' : (
      'base': (
        'font-size-base':    1rem,
      ),
    ),
  ),
);

```

Custom fonts

First, create a specific SCSS file to declare your custom font(s).

```
'website_airproof/static/src/scss/fonts.scss',
],
},
```

Then, use the `@font-face` rule to allow you custom font(s) to be loaded on your website.

/website_airproof/static/src/scss/fonts.scss

```
@font-face {
  font-family: "My Custom Font", Helvetica, Helvetica Neue;
  font-weight: 400;
  font-style: normal;
  src: url('/fonts/my-custom-font.woff') format('woff'),
       url('/fonts/my-custom-font.woff2') format('woff2');
}
```

/website_airproof/static/src/scss/primary_variables.scss

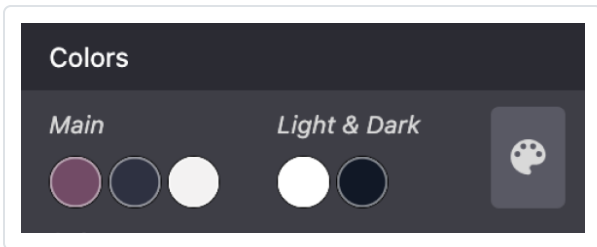
```
$o-theme-font-configs: (
  'Proxima Nova': (
    'family': ('Proxima Nova', sans-serif),
    'properties': (
      'base': (
        'font-size-base': 1rem,
      ),
    ),
  ),
);
```

Tip

It is recommended to use the `.woff` and/or `.woff2` format for your fonts.

Colors

The Website Builder relies on palettes composed of five named colors. Defining those in your theme ensures it stays consistent.

o-color-2 Secondary**o-color-3** Extra (Light)**o-color-4** Whitish**o-color-5** Blackish

Declaration

```
/website_airproof/static/src/scss/primary_variables.scss
```

```
$o-color-palettes: map-merge($o-color-palettes,
(
  'airproof': (
    'o-color-1': #bedb39,
    'o-color-2': #2c3e50,
    'o-color-3': #f2f2f2,
    'o-color-4': #ffffff,
    'o-color-5': #000000,
  ),
),
);
```

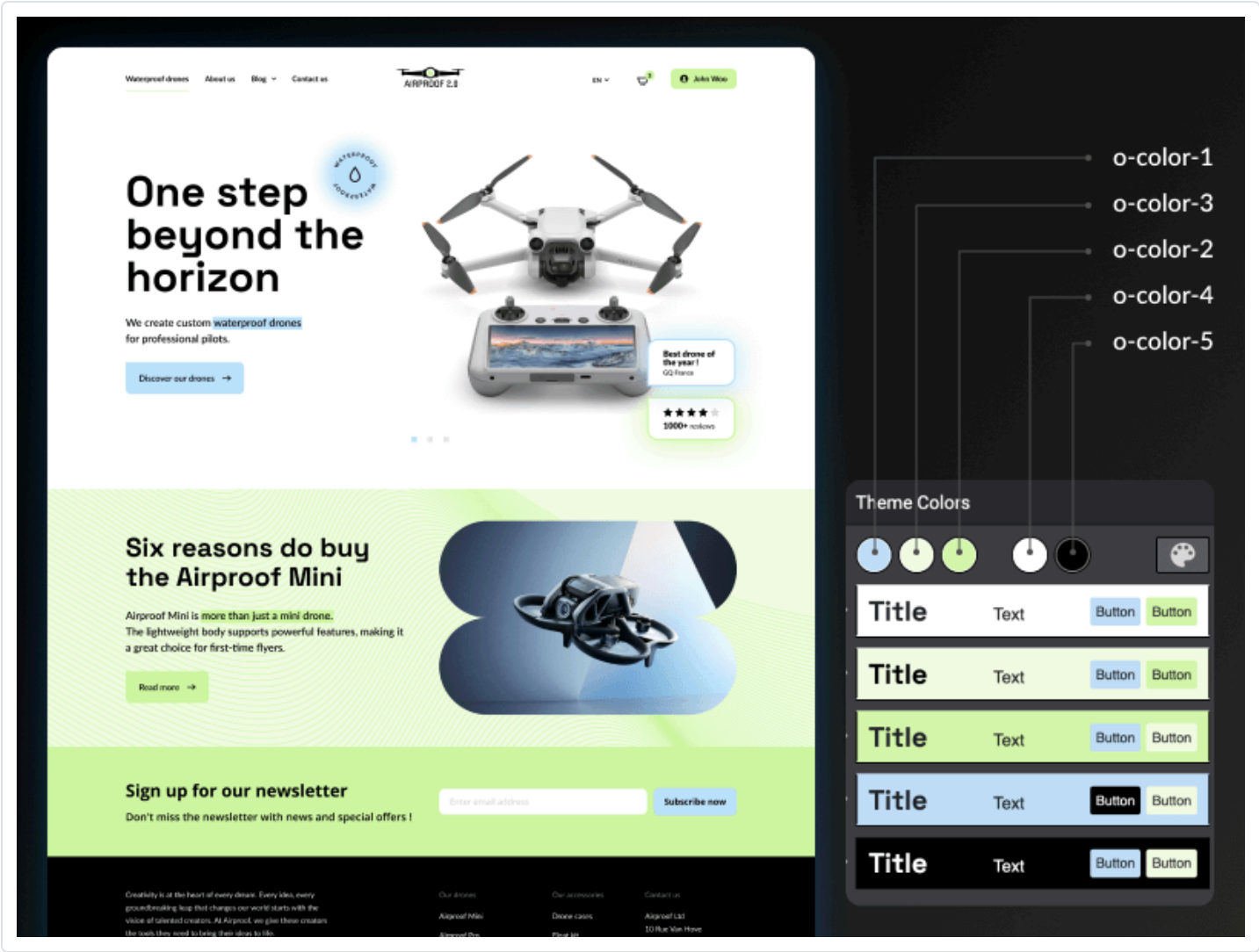
Add the created palette to the list of palettes offered by the Website Builder.

```
$o-selected-color-palettes-names: append($o-selected-color-palettes-names, 'airproof',
```

Use

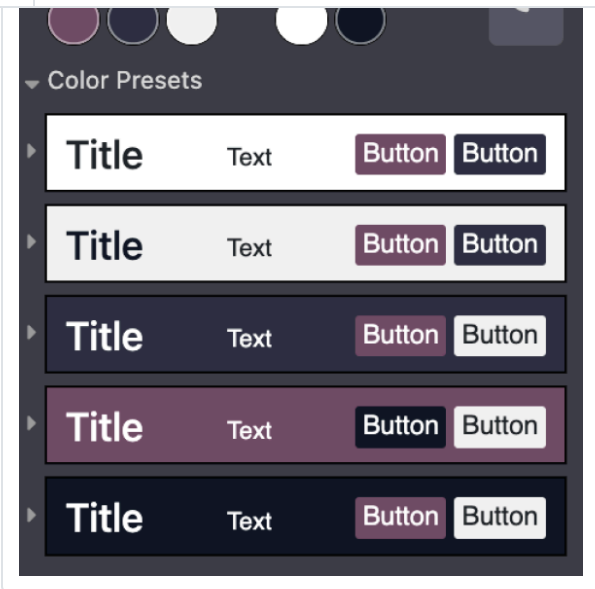
```
/website_airproof/static/src/scss/primary_variables.scss
```

```
),  
);
```



Color combinations

Based on the previously defined five color palettes, the Website Builder automatically generates five color combinations, each defining a color for the background, text, headings, links, primary buttons, and secondary buttons. These colors can be customized later by the user.



The colors used in a color combination are accessible and can be overridden through `$o-color-palettes` using a specific prefix (`o-cc` for color combination).

`/website_airproof/static/src/scss/primary_variables.scss`

```
$o-color-palettes: map-merge($o-color-palettes,
(
  'airproof': (
    'o-cc*-bg': 'o-color-*',
    'o-cc*-text': 'o-color-*',
    'o-cc*-headings': 'o-color-*',
    'o-cc*-h2': 'o-color-*',
    'o-cc*-h3': 'o-color-*',
    'o-cc*-h4': 'o-color-*',
    'o-cc*-h5': 'o-color-*',
    'o-cc*-h6': 'o-color-*',
    'o-cc*-link': 'o-color-*',
    'o-cc*-btn-primary': 'o-color-*',
    'o-cc*-btn-primary-border': 'o-color-*',
    'o-cc*-btn-secondary': 'o-color-*',
    'o-cc*-btn-secondary-border': 'o-color-*',
  ),
),
);
```

combination.

The default text color is `o-color-5`. If the background is too dark, it will automatically change to the `o-color-4` color.

See also

[Color combinations SCSS](#)

Demo page

The Website Builder automatically generates a page to view the color combinations of the theme color palette: <http://localhost:8069/website/demo/color-combinations>

Gradients

You can also define gradients for the menu, header, footer and copyright bar, directly in your `primary_variables.scss` file.

Declaration

/website_airproof/static/src/scss/primary_variables.scss

```
$o-website-values-palettes: (  
  (  
    'menu-gradient': linear-gradient(135deg, rgb(203, 94,  
    'header-boxed-gradient': [your-gradient],  
    'footer-gradient': [your-gradient],  
    'copyright-gradient': [your-gradient],  
  ),  
);
```

Bootstrap variables

Odoo includes Bootstrap by default. You can use all variables and mixins of the framework.

If Odoo does not provide the variable you are looking for, there could be a Bootstrap variable that allows it. Indeed all Odoo layouts respect Bootstrap structures and use Bootstrap

and *not* the `primary_variables.scss` file.

Declaration

/website_airproof/__manifest__.py

```
'assets': {  
    'web._assets_frontend_helpers': [  
        ('prepend', 'website_airproof/static/src/scss/bootstrap_
```

Use

/website_airproof/static/src/scss/bootstrap_overridden.scss

```
// Typography  
$h1-font-size: 4rem !default;  
  
// Navbar  
$navbar-nav-link-padding-x: 1rem !default;  
  
// Buttons + Forms  
$input-placeholder-color: o-color('o-color-1') !default;  
  
// Cards  
$card-border-width: 0 !default;
```

Tip

That file must only contain definitions and overrides of SCSS variables and mixins.

Warning

Don't override Bootstrap variables that depend on Odoo variables. Otherwise, you might break the possibility for the user to customize them using the Website Builder.

See also

[Bootstrap overridden SCSS](#)

Demo page

<http://localhost:8069/website/demo/bootstrap>

Font sizes

Odoo has CSS font size classes to separate style (font sizes) and semantic (tag and styles in general). Both logic can be combined to be more flexible.

See also

[Bootstrap documentation about display headings](#)

Text style

Odoo's Website Builder allows to select a style for your text. Some are just tag related like the `Header` with no extra CSS class. Other combine tag and style directly applied on them like the `Header 1 Display`.

Header 1 Display 1

Header 1 Display 2

Header 1 Display 3

Header 1 Display 4

Header 1

Header 2

Header 3

Header 4

Header 5

Header 6

Paragraph

Light

Small

Code

Quote

```

<!-- h1 with display heading sizes -->
<h1 class="display-1">Heading 1 with Display Heading 1 size</h1>
<h1 class="display-2">Heading 1 with Display Heading 2 size</h1>
<h1 class="display-3">Heading 1 with Display Heading 3 size</h1>
<h1 class="display-4">Heading 1 with Display Heading 4 size</h1>

<!-- Lead text - named "Light" in the dropdown -->
<p class="lead">A text typically used as an introduction.</p>

<!-- Small text -->
<p class="o_small">Body text with a smaller size.</p>

```

Sizing classes

The sizing classes are added on a freshly created `span` tag inside the targeted element (See examples below).

14	Small
16	Normal
20	Heading 5
24	Heading 4
32	Heading 3
40	Heading 2
48	Heading 1
56	Display 4
64	Display 3
72	Display 2
80	Display 1

Heading and body text

Assuming these classes can be applied on any text element, take `h2` as an example below:

```
<!-- h2 sized like an h1 -->
<h2><span class="h1-fs">Heading</span></h2>

<!-- h2 sized with other heading sizes -->
<h2><span class="h2-fs">Heading</span></h2>
<h2><span class="h3-fs">Heading</span></h2>
<h2><span class="h4-fs">Heading</span></h2>
<h2><span class="h5-fs">Heading</span></h2>
<h2><span class="h6-fs">Heading</span></h2>

<!-- h2 sized like a normal paragraph (base size, 16px by default) -->
<h2><span class="base-fs">Heading</span></h2>

<!-- h2 sized like a small text (14px by default) -->
<h2><span class="o_small-fs">Heading</span></h2>
```

Display headings

If larger titles are needed, Odoo uses Bootstrap's Display Headings classes, from `display-1` to `6`.


```
<h2><span class="display-4 fs">Heading</span></h2>
```

Note

The Website Builder allows the user to configure only sizes from `Display 1` to `Display 4`. You can set the other sizes (`5` and `6`) to use them in your code but the user won't be able to modify them directly within the Website Builder interface.

Website settings

Global options related to the website can be set through the website record by following the structure below.

Declaration

```
/website_airproof/data/website.xml
```

```
<?xml version="1.0" encoding="utf-8"?>
<odoo noupdate="1">
  <record id="website.default_website" model="website">
    <field name="name">Airproof</field>
    <field name="logo" type="base64" file="website_airproof/logo.png"></field>
    <field name="favicon" type="base64" file="website_airproof/favicon.png"></field>
    <field name="shop_ppg">18</field>
    <field name="shop_ppr">3</field>
    <field name="shop_gap">16px</field>
    <field name="shop_opt_products_design_classes">
      o_wsale_products_opt_layout_catalog
      o_wsale_products_opt_design_cards
      o_wsale_products_opt_name_color_regular
      o_wsale_products_opt_thumb_cover
      o_wsale_products_opt_img_hover_zoom_out_light
      o_wsale_products_opt_has_cta
      o_wsale_products_opt_actions_onhover
      o_wsale_products_opt_wishlist_fixed
      o_wsale_products_opt_cc
      o_wsale_products_opt_cc1
      o_wsale_products_opt_rounded_2
      o_wsale_products_opt_actions_promote
      o_wsale_products_opt_actions_subtle
    </field>
  </record>
</odoo>
```

```
<field name="social_facebook">https://www.facebook.com</field>
<field name="social_instagram">https://www.instagram.com</field>
<field name="social_linkedin">https://www.linkedin.com</field>
<field name="social_youtube">https://www.youtube.com/</field>
</record>
</odoo>
```

See also

Website model [↗](#) from the eCommerce App:

- Fields prefixed with `shop_` are related to the `/shop` page configuration;
- Fields prefixed with `product_page_` are related to the product page configuration;
- “Cards Design” classes shared between `/shop` page and dynamic products snippet.

Field	Description
name	Name of the website (displayed in the browser)
logo	Path to the logo (previously created into a record)
favicon	Path to the favicon (previously created into a record)
shop_ppg	Number of products shown per page in the e-commerce
shop_ppr	Number of products shown per rows (in a page) in the e-commerce
shop_gap	Gutter value used to separate items of the product grid
shop_opt_products_design_classes	List of CSS classes related to the Product Design options.
cookies_bar	Enable/disable the cookies bar
contact_us_button_url	URL of the Contact us page (For instance, used in the standard header templates).

social_instagram	URL of a Instagram profile
social_linkedin	URL of a LinkedIn company profile
social_youtube	URL of a YouTube channel

Note

`website.default_website` is the default reference whenever you work with only one website. If there are several websites in your database, this record will refer to the default site (the first one).

Views

For some options, in addition to the Website Builder variable, you also have to activate a specific view.

By reading the source code, templates related to options are easily found.

```
<BuilderContext action="'websiteConfig'">
  <BuilderRow label.translate="...">
    <BuilderCheckbox actionParam="{views: ['website.show_website_info']}" />
  </BuilderRow>
</BuilderContext>
```

```
<template id="..." inherit_id="..." name="..." active="True"/>
<template id="..." inherit_id="..." name="..." active="False"/>
```

See also

In such cases, the `<template id="...">` syntax should be preferred over the `<record id="..." model="ir.ui.view">` syntax when defining the record in XML.

See **template**, especially about the `active` value.

Presets

```
/website_airproof/data/presets.xml
```

```
<record id="module.view" model="ir.ui.view">
  <field name="active" eval="False"/>
</record>
```

Example

Changing the menu items' horizontal alignment

```
/website_airproof/data/presets.xml
```

```
<record id="website.template_header_default_align_center" m
  <field name="active" eval="True"/>
</record>
```

The same logic can be used for others Odoo apps as well.

eCommerce - Display products categories

```
/website_airproof/data/presets.xml
```

```
<record id="website_sale.products_categories" model="ir.ui
  <field name="active" eval="False"/>
</record>
```

Portal - Disable the language selector

```
/website_airproof/data/presets.xml
```

```
<record id="portal.footer_language_selector" model="ir.ui.v
  <field name="active" eval="False"/>
</record>
```

bundle specifies the list of assets loaded on the website frontend (public pages), and the goal is to add your SCSS and JS files to the bundle. The Website Builder uses its own bundles (for example: `website.website_builder_assets` and `website.assets_wysiwyg`).

This is a non-exhaustive list of the frequently used bundles for a website:

Bundle	Description
web._assets_primary_variables	Mainly used for the <code>primary_variables.scss</code> file
web._assets_secondary_variables	Mainly used for the <code>secondary_variables.scss</code> file
web._assets_frontend_helpers	Mainly used for the <code>bootstrap_overridden.scss</code> file
web.assets_frontend	You can add all your custom SCSS, JS or QWeb JS files
website.website_builder_assets	Add your JS files related to the Website Builder options behaviors (for instance, a custom method for your custom building block)
web._assets_bootstrap_frontend	If you need to extend Bootstrap through the Bootstrap Utilities API, for example

Styles

The Website Builder together with Bootstrap are great for defining the basic styles of your website. But to design something unique, you should go a step further. For this, you can easily add any SCSS file to your theme.

Declaration

```
/website_airproof/__manifest__.py
```

```
'assets': {  
    'web.assets_frontend': [  
        'website_airproof/static/src/scss/theme.scss',  
    ],  
},
```

Example

```
/website_airproof/static/src/scss/theme.scss
```

```
blockquote {  
  border-radius: $rounded-pill;  
  color: o-color('o-color-3');  
  font-family: o-website-value('headings-font');  
}
```

Interactivity

Odoo supports three different kinds of JavaScript files:

- **plain JavaScript files** (no module system),
- **native JavaScript module**, and
- **Odoo modules** (using a custom module system).

Most new Odoo JavaScript codes should use the native JavaScript module system. It's simpler and brings the benefit of a better developer experience with better integration with the IDE.

Declaration

```
/website_airproof/__manifest__.py
```

```
'assets': {  
  'web.assets_frontend': [  
    'website_airproof/static/src/js/theme.js',  
  ],  
},
```

Note

If you want to include files from an external library, you can add them to the `/lib` folder of your module.

Tip

- Variables and functions should be *camelCased* (`myVariable`) instead of *snake_cased* (`my_variable`).
- JavaScript classes use *PascalCase* (`MyClass`).
- Do not name a variable `event`; use `ev` instead. This is to avoid bugs on non-Chrome browsers, as Chrome is magically assigning a global `event` variable (so if you use the `event` variable without declaring it, it will work fine on Chrome but crash on every other browser).
- Use strict comparisons (`===` instead of `==`).
- Use double quotes for all strings (such as `"Hello"`, `".x_nav_item"`).
- If you're using native standard JS functions (`start()`, `willStart()`, `cleanForSave()`, etc), make sure you call the parent implementation (for example, `super.start(...arguments)`; check if it's necessary in the standard code).

See also

- [Odoo JavaScript coding guidelines](#)
- [Overview of the Odoo JavaScript framework](#)
- [Odoo Experience Talk: Best practices to use the Website Builder like a professional designer](#)

Get Help

[Contact Support](#)[Ask the Odoo Community](#)